Building a Real-Time AI-Driven Alert & Response System – ‘RADAR’

To Support Disease Outbreak Management in Kenya

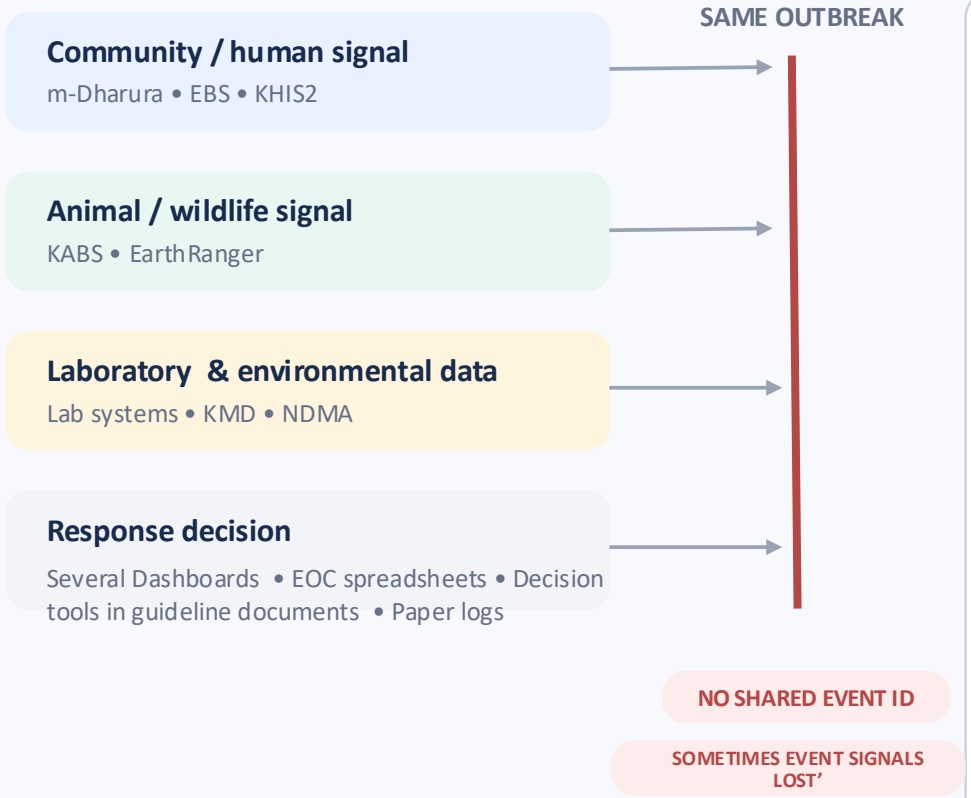


Washington State University · CGP · KNPHI · KEMRI · University of Michigan . KIPRE

A ONE HEALTH COLLABORATION

The core problem: Kenya is data-rich, but outbreak events remain fragmented

Signals are captured across many systems, but the same event does not yet have a shared identity across the surveillance-to-response pathway.



Operational consequences

1	**Manual reconstruction**	Teams must reconcile the same event across systems before they can act.
2	**Detection delinked from response**	Escalation and response decisions are delayed and inconsistently documented.
3	**7-1-7 cannot be measured continuously**	Critical timestamps sit in different systems and are reconstructed retrospectively.
4	**Integration becomes expensive if attempted all at once**	Heterogeneous architectures, rules and data volumes make "integrate everything" brittle and difficult to sustain.

Design implication: integrate incrementally around high-value events — not by rebuilding every source system.

Source: RADAR Investment Document (revised, 2026)

Minimum Event Record (MER): the practical bridge to interoperability

A single event identity that links fragmented signals across climate, animal, human, and operational systems

1 Why MER is needed

Many systems, valuable signals — but no common event identity

Climate & weather signals
KMD projections · heavy rainfall · flooding
Different formats, models and identifiers
ID A-104

Environmental / vector signals
mosquito abundance / breeding conditions
Often no single routine system
ENV-331

Animal health signals
KABS · animal illness · abortions · deaths
Different case IDs and reporting systems
Case #KABS-22

Human health signals
EBS · KHIS / DHIS2 · EMRs · laboratory systems · LMIS · human illness · deaths
Multiple systems, different identifiers
Alert EBS-7

Other sources (supporting)
community reports · field investigations · media / event signals
Report M-18

Without a common event identity, signals remain disconnected and detection-to-decision is delayed.
One missed link can mean a missed outbreak.

2 The Minimum Event Record (MER)

A simple, shared event record that connects the dots

Minimum Event Record (MER)

A minimum necessary, standards-based event record that links related signals from different systems without replacing source systems.

- **Event ID** — A single identity across systems
- **Source system** — Where each signal originated
- **Time / date** — When the event occurred
- **Linked records** — Related IDs from source systems
- **Location** — Where the event is occurring
- **Severity indicators** — Signal strength / potential impact
- **Hazard / syndrome** — What is happening
- **Status / escalation** — e.g., new, under investigation, escalated
- **Species / population affected** — Animals, humans or both
- **Confidence / verification** — e.g. unverified, probable, confirmed

MER complements, not replaces, source systems.

Supports interoperable data fusion and disease-agnostic event tracking.

LINK FRAGMENTED SIGNALS INTO ONE EVENT

3 Example: how an RVF event can be missed — and how MER helps

Different systems see parts of the picture, but MER links them.

1. **KMD projects heavy rainfall and flooding** — Climate signal suggests increased risk — KMD ID: **A-104**
2. **Mosquito breeding increases** — Environmental / vector signal (often no single routine system) — Vector ID: **ENV-331**
3. **Livestock abortions / deaths reported in KABS** — Animal health signal — KABS ID: **Case #KABS-22**
4. **Human illness / deaths appear in EBS + KHIS / EMRs + laboratories / LMIS** — Human health signals — EBS ID: **Alert EBS-7** · **Lab Ref: 83**

ONE LINKED EVENT IN MER — MER Event ID: **RVF-ISI-2026-001**

MER links climate, environmental, animal, and human signals into one event thread, enabling earlier recognition, DMT-PHE decision support, and coordinated response.

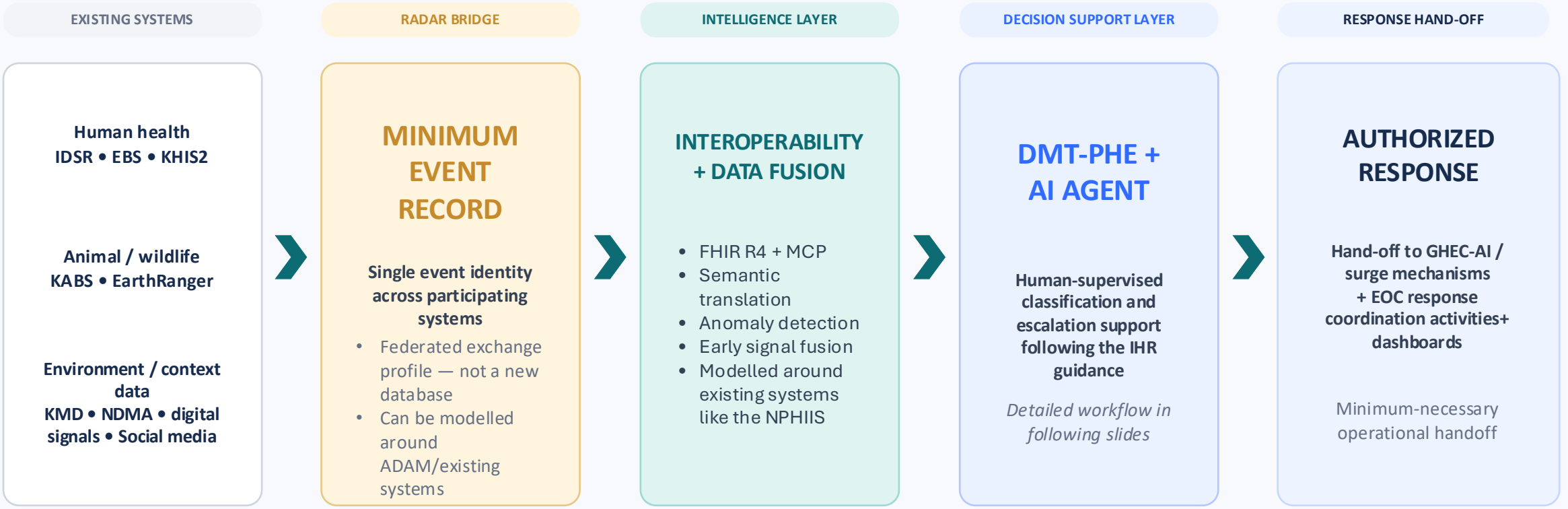
From fragmented signals to one event thread → faster recognition, clearer escalation, and more timely response

PEOPLE
ANIMALS
ENVIRONMENTS
SAFER TOGETHER

Source: RADAR Investment Document (revised, 2026)

RADAR: an operational integration and decision-intelligence layer — not another database

Embedded within existing EOC workflows; designed to complement, not replace, Kenya's surveillance infrastructure.



DESIGN PRINCIPLES

COMPLEMENT, NOT REPLACE | OPEN STANDARDS | INCREMENTAL / MODULAR | GOVERNMENT STEWARDSHIP | EOC-EMBEDDED

Source: RADAR Investment Document (revised, 2026)

RADAR Approach to Integration/Interoperability

Why "Integrate Everything" is a challenge

- Too many systems, datasets and architectures to connect at once
- Legacy platforms carry different standards, topologies and business rules. Some systems may lag if not maintained routinely
- High computing, infrastructure and maintenance costs needed can quickly become unsustainable
- Complexity can outweigh value if integration produces limited additional operational benefit
- Large, all-at-once solutions risk becoming brittle and difficult to maintain

RADAR approach emphasizes 4 things:

- Build on existing national systems rather than parallel platforms. No new interfaces/embed everything
- Slice the problem into manageable, high-value use cases → demonstrate value → expand incrementally, and sustain gains in a nimble platform
- Leveraging technologies such as AI to show show incremental value
- Design integration around government ownership, operational workflows and measurable improvements in detection-to-response.

Decision- Making Tool for Public Health Emergencies (DMT-PHE) in Kenya

Institutionalizing AI-Supported Decision Architecture for Public Health Emergencies

REPUBLIC OF KENYA | NPHI | CGP CENTER FOR GLOBAL HEALTH AND PANDEMIC INTELLIGENCE | TDDAP2 Strengthening health systems to protect against public health threats | WASHINGTON STATE UNIVERSITY | Kenya Institute of Primate Research | MICHIGAN MEDICINE UNIVERSITY OF MICHIGAN

Kenya's Public Health Surveillance & Response: Gaps

- **Recurring outbreaks/PHEICS –** Frequent epidemics (Cholera, Mpox, RVF, anthrax, Kala azar, measles), COVID-19, VHF threats exposed *fragmented decision-making processes* and delayed escalation at both county and national levels.

- **Uneven county readiness –** Devolution created wide disparities in outbreak management capacity, weak coordination, and access to surge resources.

- **Accountability and coordination gaps:** JEE 2024 and SPAR 2024 highlighted the absence of a standardized framework for timely decision-making and response to PHE

- **Need for a unified decision system –** A national tool is essential to align counties, KNPHI, and partners under *common thresholds*, timelines, and support mechanisms to meet the 7-1-7 targets.

- **Workforce turnover and Knowledge to practice-** Staff rotation, attrition,infrequent events limit consistent realtime use of PHE protocols

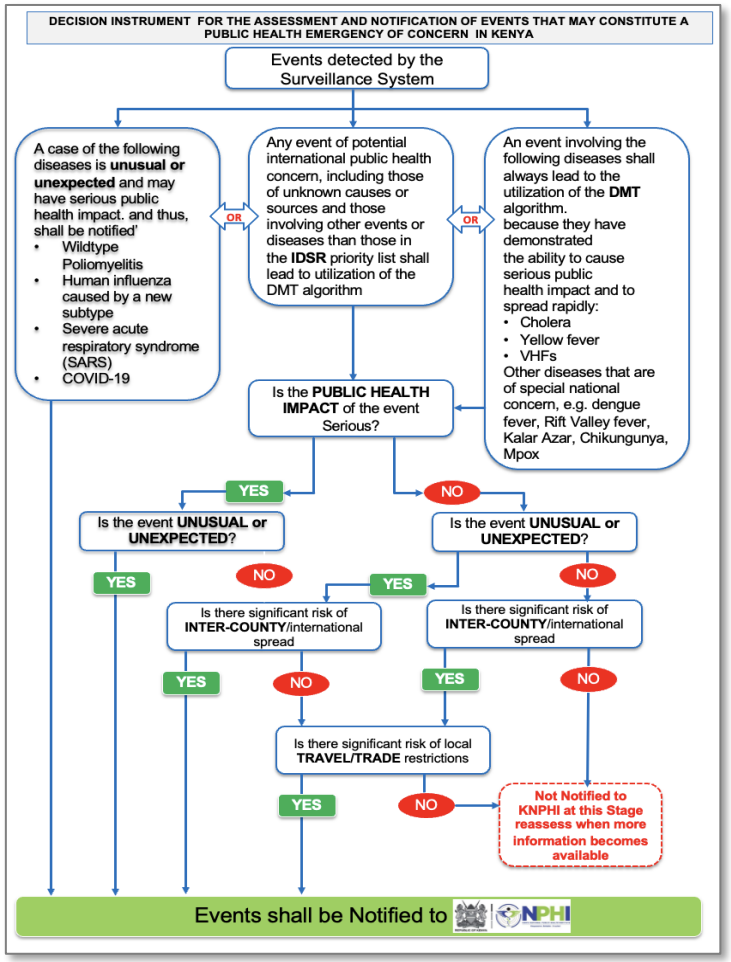


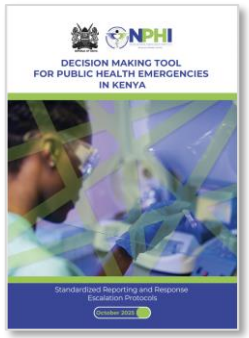


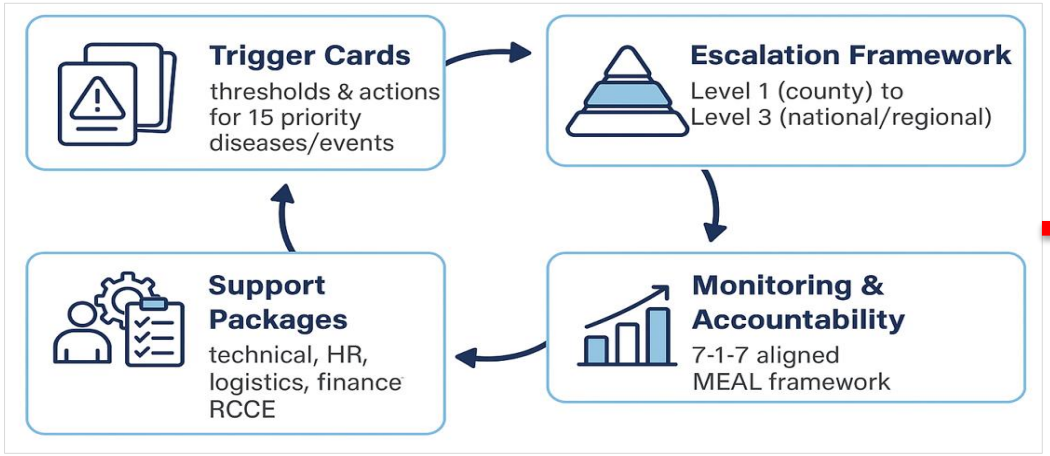




WHO (IHR 2005 Annex 2) Adaptation and DMT-PHE: Trigger Card (22 ETC)



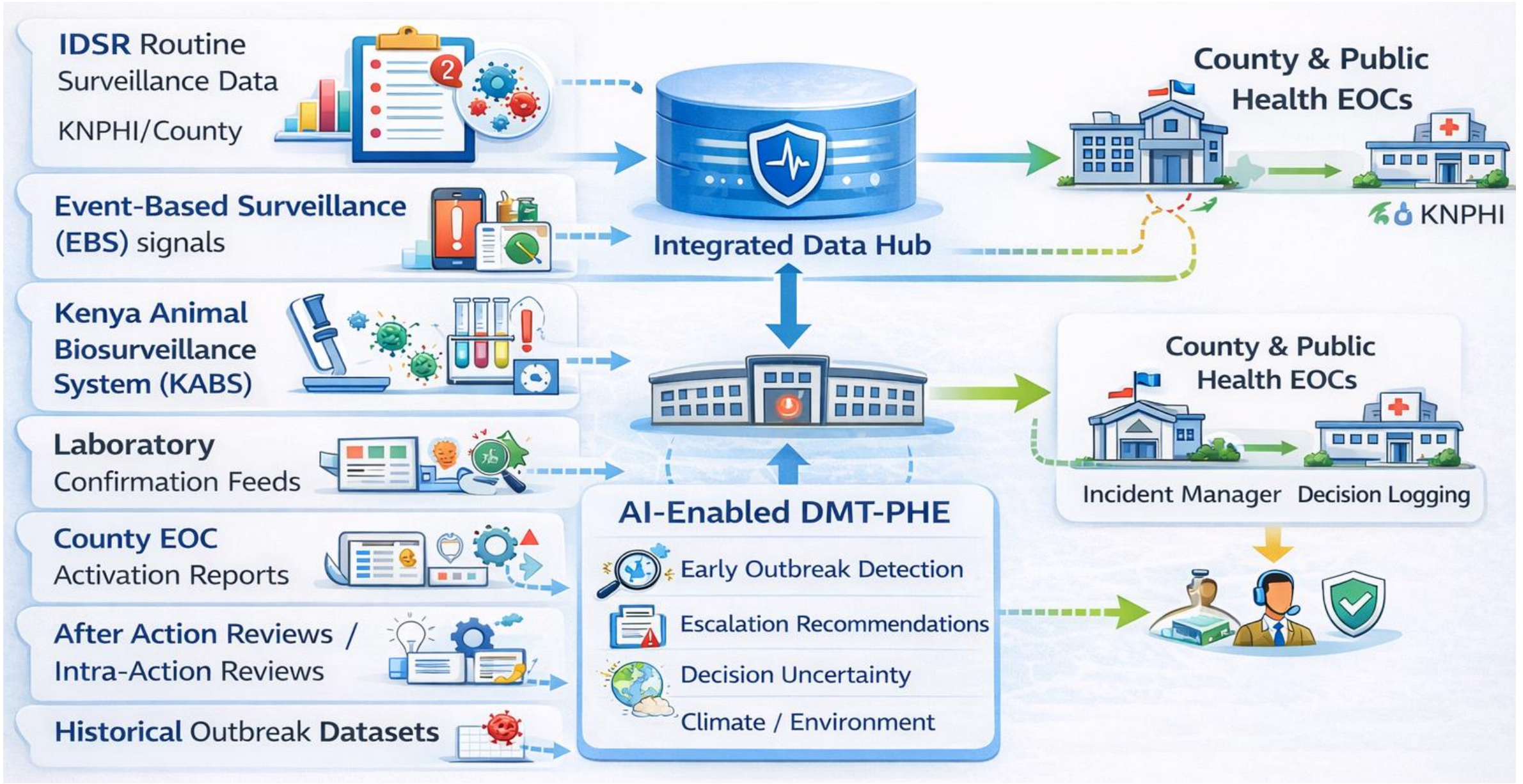




Ebola

Priority Disease / Event	Case Definitions & Outbreak Definition	Alert Thresholds (Early Signals)	Escalation Triggers (Tiered – 3 levels)	Required Actions & Timelines (7-1-7)	Support Package (Technical/HR/ Logistics/Finance /RCCE)	Responsible Actors	Monitoring & De-escalation / Exit Criteria
Filoviruses (Ebola Virus Disease / Marburg Virus Disease)	Suspected: Any person with acute fever ≥38°C and any of the following: severe headache, muscle pain, vomiting, diarrhea, abdominal pain, or unexplained bleeding. Probable: A suspected case with an epidemiological link to a confirmed case or death consistent with EVD/MVD, or any unexplained community death with bleeding. Confirmed: Laboratory confirmation of Ebola or Marburg virus infection by RT-PCR or serology. Outbreak Definition: One laboratory-confirmed case of Ebola or Marburg Virus Disease.	• Any single case meeting the suspected case definition. • Unexplained death with hemorrhagic signs. • Clusters of febrile illness with bleeding or sudden deaths among health workers or household contacts.	Level 1- Subcounty/County Response: Single suspected case or death with bleeding symptoms, managed locally with isolation and basic IPC capacity. Level 2 - County/National Response: Two or more epidemiologically linked suspected cases, or cluster among health workers, requiring County RRT deployment and testing at KEMRI/NPHL. Level 3 - National/Regional Surge: Laboratory-confirmed case(s), multi-county spread, or cross-border risk, triggering national activation and international notification.	≤ 24 h – Isolate suspected case, notify KNPHI PHEOC and County IMS, initiate contact listing, and collect blood for RT-PCR. ≤ 72 h – Deploy County RRT (Clinical, Epi, Lab, IPC, RCCE); enhance IPC; initiate safe burial arrangements; begin community risk communication. ≤ 7 d- Confirm diagnosis; expand contact tracing, monitoring, and follow-up; conduct situation analysis and daily SITREP updates; coordinate partner response.	Technical: Epidemiologists, clinicians, laboratory scientists, IPC officers, burial teams, RCCE staff. HR: County RRT, sub-county teams, community health promoters, and contact tracers. Logistics: PPE, body bags, disinfectants, ambulances, specimen transport kits, isolation space. Finance: Rapid emergency funds for RRT deployment, training, and supplies. RCCE: Continuous engagement with communities on safe burials, IPC, and stigma reduction.	Sub-County HMT / CHMT- Case detection, isolation, notification, and initial contact tracing. COHU + CHPs- Community sensitization, rumor tracking, and early alerts. KNPHI PHEOC- National coordination, activation of nationalRRT, and situation reporting. KNPHI EPR- Technical oversight, analytics, and international liaison. KEMRI / NPHL- RT-PCR testing and result confirmation. NEMA- Waste management and decontamination oversight in affected facilities. NDOC / DRM- Multisect or logistics coordination and partner engagement. Security Agencies- Security, movement control, and support to safe burial teams. Partners (WHO, AFENET, KRCS, Amref, etc.) – Surge staffing, technical support, RCCE, and training.	Indicators: Case fatality rate, % contacts traced within 48 h, # HCWs infected, time to isolation and notification. De-escalation / Exit: No new confirmed cases for 42 days (two incubation periods) after last case tests negative and all contacts complete follow-up.

Core Data Streams: Integrated Surveillance feeds into DMT for PHE



Kenya DMT-PHE AI Assistant: Evaluation, Pilot Progress (7-1-7 SIMEX)

Kenya DMT-PHE Ai Assistant: https://dmt-agent.streamlit.app



Pilot Evaluation: Case studies detection, escalation and response delays

Mpox

Early Cluster Detection



2+ sub-counties report **lesions & fever** cases

AI detects abnormal **signal aggregation**

Cross-checks historical & **geographic data**

Recommends:

Preliminary Outbreak Classification

Activate County Rapid Response Team

Recommends:

Preliminary Outbreak Classification

CLASSIFY & ESCALATE

PHEIUE

Escalation Delay Detection



Lab confirms case

No escalation within **48 hours**

AI detects **delay**

24h 48h

Triggers:

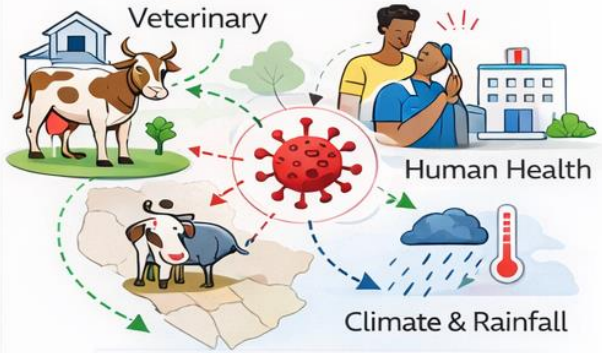
Supervisory Alert

Decision Latency Breach

IMPROVE TIMELINESS

Rift Valley Fever (RVF)

One Health Zoonotic Alert



Livestock abortions

Febrile illness cases

Heavy rainfall anomaly

Recommends:

One Health Investigation

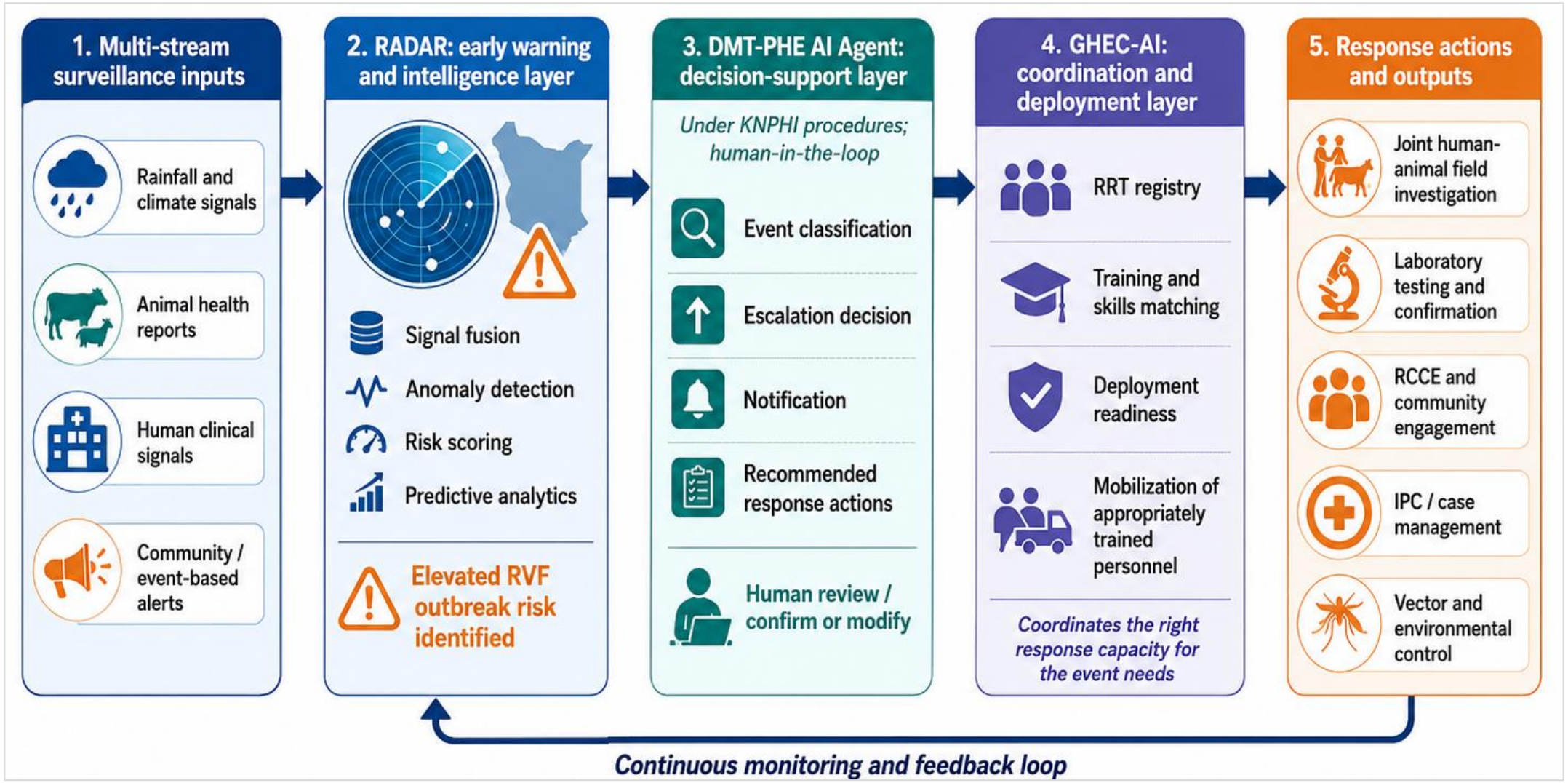
Joint Human + Animal Response

Cross-Sector Coordination

ONE HEALTH RESPONSE

Complementarity DMT-PHE AI Agent “RADAR” and GHEC-AI

Zoonotic outbreak scenario: suspected Rift Valley fever outbreak in Northern Kenya



Phased RADAR Implementation · MVP → Deployment → Scale-Up (3 Years)

Mths 1–6	Mths 7–18	Year 2	Year 3

Phase 0

Minimum Viable Product (MVP) *Months 1–6*

- Agree and code the shared Minimum Event Record for priority surveillance events
- Map and establish initial interoperability/data flows for priority source systems
- Demonstrate integrated DMT-PHE signal-ingestion, classification, escalation & confirmation
- Define RADAR / DMT-PHE / GHEC-AI interface specifications jointly with Sobek

🔒 Gate 0 · Technical & operational MVP go/no-go — before broader national/county deployment

Phase 1

Platform Preparation & National EOC Deployment *Months 7–18*

- Complete dataset inventory, priority system connectors & outbreak userguides (4 categories)
- Deploy digitized DMT-PHE at national EOC; integrate with GHEC-AI workflow
- Conduct 8-week parallel validation deployment generating concordance dataset
- Finalize Technical Product Profile (TPP) with product hardening and security controls

🔒 Gate 1 · AI-expert concordance ≥0.85 · SUS ≥75 · Completeness ≥80% · Zero adverse events · Minimum event record published

Phase 2

County EOC Deployment & Scalability Testing *Year 1–2*

- Deploy RADAR & DMT-PHE at 4 priority county EOCs; product hardening & load testing
- County staff training, parallel validation, AI model retraining using operational data
- Cybersecurity review, scalability and interface validation in KNPHI/DHA infrastructure
- Evaluate RADAR / DMT-PHE / GHEC-AI handoff for timeliness, reliability & workflow fit

🔒 Gate 2 · >80% dataset integration · Concordance ≥0.88 · Detection-to-classification time reduced · SUS ≥78 all sites

Phase 3

Evaluation, Evidence Generation & KNPHI Transition *Year 2–3*

- Prospective evaluation; endline assessment; 7-1-7 compliance and AI performance review
- Operational handover to KNPHI Public Health Intelligence Directorate
- Scale-up evidence report, policy brief, dissemination via Africa CDC & WHO PDX
- Peer-reviewed publications documenting system performance and implementation lessons

🔒 Gate 3 · Concordance ≥0.90 · SUS ≥80 · False-negative rate below ceiling · KNPHI budget pathway · 2 publications

RADAR partnership model

Distinct technical workstreams, anchored by KNPHI/DHA ownership, standards, data governance and operational adoption.

Government stewardship & digital architecture

KNPHI

- Government lead, system custodian and primary operational user
- Data/system access, DMT-PHE stewardship and EOC/county workflow integration
- Operational transition, adoption and institutionalization

DHA

DIGITAL HEALTH AGENCY

- National digital architecture, interoperability standards and FHIR alignment
- Hosting, cybersecurity, data-residency and production deployment guidance
- Alignment/certification within Kenya's digital health ecosystem

Prime & RADAR integration support

- Scientific/technical coordination and grant oversight
- MER, data-fusion layer, connectors, APIs/MCP and user-facing RADAR
- TPP/product coordination and formal evaluation leadership

AI-enabled DMT-PHE lead

- Agentic decision-support architecture and integration
- Benchmarking, safety/guardrails and responsible AI validation
- Containerization, interfaces and ongoing technical maintenance

Deployment & operationalization

- DMT-PHE national/county rollout and EOC workflow integration
- Training, simulation, mentorship and user-support/helpdesk
- Adoption monitoring and scale-up implementation lessons

Evaluation & implementation science

- Epidemiology, ethics/research governance and implementation science
- Baseline/endline evaluation, data-quality assessment and validation
- Analysis of detection-to-decision, usability and 7-1-7 outcomes

One Health integration & Stakeholder engagement

- Animal, wildlife and environmental surveillance linkages
- Cross-sector interoperability and One Health workflow validation
- Stakeholder/policy engagement and scale-up learning

Enterprise product-development support (TBD)

To be selected early: product engineering • UI/UX • scalability/load testing • cybersecurity • production hardening • deployment support

Source: RADAR Investment Document, revised September 2026